Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Levine, Howard G Ph.D.		
Project Title:	Water Offset Nutrient Delivery ExperRment (WONDER)		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	321-861-3502
Organization Name:	Space Life Sciences Laboratory		
PI Address 1:	Mail Code YA-E4-B		
PI Address 2:			
PI Web Page:	http://lssc.ksc.nasa.gov/		
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32899	**Congressional District:**	24
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	07/31/1999	**End Date:**	03/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	In development, no current flight assignment		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dreschel, Thomas (NASA)		
Grant/Contract No.:	NCC10-52		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	There is a need for microgravity-based plant culture nutrient delivery systems (NDS') for both bioregenerative Advanced Life Support and plant research functions. The provision of adequate levels of water (without causing water logging) and oxygen to the root zone are the most crucial components deterring major advances in this area. The dominance of the surface tension of water under microgravity conditions has often been found to lead to either severe water-logging or excessive drying in the root zone. Consequently, differences in plant growth responses between spaceflight experiments and their ground controls are expected based merely upon differences in moisture distribution patterns between the two conditions. This project will address the question of "comparability of environmental conditions" between the spaceflight and ground control experiments for both a porous tube plant NDS and a substrate-based NDS by employing 3 different wetness level treatments for both of these approaches. It is anticipated that different pre-set wetness levels		

	than those used on Earth will be required to support optimal plant growth in space. Dry seeds will be initiated 7 days prior to launch within each of the 6 experimental treatments (3 porous tubes plus 3 substrate compartments). There will be 15 wheat (Triticum aestivum cv Apogee) plants within each treatment. Once on-orbit, the experimental wetness level treatments will be initiated by the crew. Time-lapsed video recording of the plants will monitor growth over time. At recovery, the plants will be measured, and extensive tissue analyses relating to gene expression and stress-associated metabolites will be undertaken.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	If the goals of this proposal are achieved, we believe that the resulting nutrient delivery systems will have a dramatic impact on the success of future space-based plant growth efforts. This technology has the potential to reduce the mass, power, volume, thermal control requirements, consumables, and crew-time needed for the reliable growth of plants in space. It may also provide NASA with an early benefit in terms of fulfilling the requirements for a space-based vegetable production unit. Spin-off applications in the areas of home-based and agricultural hydroponic units and water conservation applications are also possible. For the latter, the porous tube approach can be used to grow plants under conditions in which water conservation is critical in as the water and nutrient solution is fed directly to the roots of the plants, thereby minimizing excess water which would be lost through evaporation.
Task Progress:	(1) Fulfilled flight project schedule tasks relating to hardware development (Critical Design Review; Phase B Review), NASA documentation deliverables (Experiment Requirements Document; Flight Safety Data Package 0/1/2), verification testing, etc. (2) Conducted growth chamber studies investigating various aspects of wheat growth within spaceflight configurations and refined protocols. (3) Conducted KC-135 investigations relating to hardware performance and operations in microgravity.
Bibliography Type:	Description: (Last Updated: 06/07/2005)
Articles in Peer-reviewed Journals	Norikane, J.H., G.K. Tynes and H.G. Levine. "Determining the Extractable Water Limit for Wheat in a Substrate-Based Media Designed for Space Flight Applications. " Applied Eng. in Agriculture, 19(5): 565-569. , Jul-2003
Articles in Peer-reviewed Journals	Norikane, J.H., S.B. Jones, S.L. Steinberg, H.G. Levine, and D. Or. "Porous Media Matric Potential and Water Content Measurements During Parabolic Flight. " Habitation, In Press. , Dec-2004
Articles in Peer-reviewed Journals	Paul, A-L., H.G. Levine, W. McLamb, K. Norwood, G. Stutte, D. Reed and R.J. Ferl. "Plant Molecular Biology in the Space station Era: Utilization of KSC Fixation Tubes with RNAlater. " Acta Astronautica, In Press. , Nov-2004
Articles in Peer-reviewed Journals	Levine, H.G. and W.C. Piastuch. "Increased Root Production in Soybeans Grown Under Space Flight Conditions. " Advances in Space Research, In Press. , Dec-2004
Articles in Peer-reviewed Journals	Anderson K., F. Ahmay, K. Burtness, C. Frazier, H.G. Levine, W. McLamb, T. Murdoch, K. Norwood, J. Rodriguez, and H. Wells. "Development of the Advanced Biological Research System (ABRS). " Advances in Space Research, In Press. , Dec-2004
Articles in Peer-reviewed Journals	Norikane, J.H., J.J. Prenger, D.T. Rouzan and H.G. Levine. "A Comparison of Soil Moisture Sensors for Space Flight Applications. " Applied Eng. In Agriculture, In Press. , Dec-2004
Papers from Meeting Proceedings	Jessica J. Prenger, J.J., S.L. Steinberg, D. Haddock, J.H. Norikane, and H.G. Levine. "Accuracy of a Point Source Thermal Soil Moisture Sensor for Space Flight Nutrient Delivery Systems. " Paper No. 2004-01-2456. International Conference on Environmental Systems. SAE 2004 Meeting. Jul-2004
Papers from Meeting Proceedings	Berkovich, Y. A., A.N. Erokhin, S.O. Smolianina, J.J. Prenger, and H.G. Levine. "Development and Testing of a Cylindrical LED Lighting Unit for a Conveyer-Type Salad Production System. " Paper No. 2004-01-2434. International Conference on Environmental Systems. SAE 2004 Meeting. Jul-2004